CS301: Design and Analysis of Algorithms - Fall 2026
Department of Computer Science
Instructor: Prof. S. Menon - smenon@university.edu
Lectures: Mon/Wed 10:00-10:50 AM, Hall C. Recitation: Thu 4 PM.

## COURSE DESCRIPTION

This course covers the design and analysis of efficient algorithms: divide and conquer, greedy methods, dynamic programming, graph algorithms, network flow, NP-completeness, and approximation algorithms. Emphasis on rigorous proofs and asymptotic analysis.
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## LEARNING OUTCOMES

Students completing this course will be able to design algorithms using standard paradigms, prove correctness, analyze running time, and recognize intractable problems. They will communicate algorithmic ideas precisely in writing.
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## GRADING POLICY

Problem sets 25%, quizzes 10%, midterm 20%, project 20%, final exam 25%.
Late submissions lose 10% per day up to three days, after which they receive zero credit. Regrade requests must be submitted within one week of grades being pos

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PROBLEM SETS

Problem Set 1 due Friday, August 21, 2026 (divide and conquer).

Problem sets are released on Gradescope at least ten days before the deadline. S
olutions must be typeset in LaTeX. Each problem set contains four to six problem
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Problem Set 2 due Friday, September 4, 2026 (greedy algorithms).

Office hours for problem set help run Tuesday through Thursday; see the course p
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 forum rather than emailing the staff.
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## QUIZZES

Quiz 1: Wednesday, September 9, 2026, 10:00-10:50 AM, in class.

Quizzes are closed-book. One handwritten sheet of notes is permitted. Quizzes emphasize definitions and short analyses rather than long proofs.

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Problem Set 3 due Friday, September 25, 2026 (dynamic programming).

The dynamic programming unit is historically the most challenging; start early. The recitation sessions in the preceding two weeks work through additional examples.

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## MIDTERM EXAM

Midterm Exam: Wednesday, October 7, 2026, 6:00-8:00 PM, Hall B.

The midterm covers all material through dynamic programming. A practice exam from a previous year will be posted one week in advance. There are no makeup exams except for documented emergencies.

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Problem Set 4 due Friday, October 23, 2026 (graph algorithms).

Graph algorithms covered include shortest paths, minimum spanning trees, and maximum flow. The programming exercise implements a maximum-flow algorithm on generated test cases.

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Quiz 2: Wednesday, October 28, 2026, 10:00-10:50 AM, in class.

COURSE PROJECT

Project proposal due Monday, November 2, 2026.

The project is completed in pairs. Choose an advanced topic (approximation, randomized, or streaming algorithms) and produce a survey plus an implementation study. The proposal is one page describing the topic, the papers you will read, and your evaluation plan.

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Problem Set 5 due Friday, November 13, 2026 (NP-completeness).

Reductions must be argued carefully: state the mapping, prove both directions, and analyze the running time of the reduction itself.

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Project presentations: Monday, November 30, 2026, 10:00 AM-12:00 PM, Hall C.

Each pair presents for eight minutes plus two minutes of questions. Attendance at all presentations is required. Slides are due to the staff the evening before.

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Final project report due Wednesday, December 9, 2026.

Reports are eight to ten pages in the provided template, excluding references. Include a reproducibility appendix describing how to run your code.

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## FINAL EXAM

Final Exam: Wednesday, December 16, 2026, 9:00 AM-12:00 PM, Hall A.

The final is cumulative with emphasis on material after the midterm. The same note-sheet policy as quizzes applies, extended to two sheets.

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## ACADEMIC INTEGRITY

All work submitted must be your own except where collaboration is explicitly permitted. Suspected violations are referred to the academic integrity board. Use of AI assistants must be disclosed in a statement accompanying each submission.

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## ACCESSIBILITY

Students needing accommodations should contact the accessibility office within the first two weeks of the semester. Extended-time arrangements for quizzes and exams are handled through the testing center.

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## APPENDIX A: WEEKLY READING SCHEDULE

Week 1: asymptotic notation and recurrences.

Required reading for week 1: textbook chapter 1, plus the posted lecture notes on asymptotic notation and recurrences. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.

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Week 2: divide and conquer.

Required reading for week 2: textbook chapter 2, plus the posted lecture notes on divide and conquer. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.

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Week 3: sorting lower bounds.

Required reading for week 3: textbook chapter 3, plus the posted lecture notes on sorting lower bounds. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.

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Week 4: greedy algorithms and exchange arguments.

Required reading for week 4: textbook chapter 4, plus the posted lecture notes on greedy algorithms and exchange arguments. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.

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Week 5: matroids.

Required reading for week 5: textbook chapter 5, plus the posted lecture notes on matroids. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.

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Week 6: dynamic programming I.

Required reading for week 6: textbook chapter 6, plus the posted lecture notes on dynamic programming I. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.

Required reading for week 6: textbook chapter 6, plus the posted lecture notes on dynamic programming I. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.

Required reading for week 6: textbook chapter 6, plus the posted lecture notes on dynamic programming I. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.

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Week 7: dynamic programming II.

Required reading for week 7: textbook chapter 7, plus the posted lecture notes on dynamic programming II. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.

Required reading for week 7: textbook chapter 7, plus the posted lecture notes on dynamic programming II. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.

Required reading for week 7: textbook chapter 7, plus the posted lecture notes on dynamic programming II. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.

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Week 8: amortized analysis.
Required reading for week 8: textbook chapter 8, plus the posted lecture notes on amortized analysis. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.
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Week 9: graph traversal and DAGs.
Required reading for week 9: textbook chapter 9, plus the posted lecture notes on graph traversal and DAGs. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.
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Week 10: shortest paths.
Required reading for week 10: textbook chapter 10, plus the posted lecture notes on shortest paths. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.
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Week 11: minimum spanning trees.

Required reading for week 11: textbook chapter 11, plus the posted lecture notes on minimum spanning trees. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.

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Week 12: maximum flow.

Required reading for week 12: textbook chapter 12, plus the posted lecture notes on maximum flow. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.

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Week 13: matchings.

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Week 14: NP-completeness I.

Required reading for week 14: textbook chapter 14, plus the posted lecture notes

on NP-completeness I. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.
Required reading for week 14: textbook chapter 14, plus the posted lecture notes on NP-completeness I. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.
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Week 15: NP-completeness II.
Required reading for week 15: textbook chapter 15, plus the posted lecture notes on NP-completeness II. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.
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Week 16: approximation algorithms.
Required reading for week 16: textbook chapter 16, plus the posted lecture notes on approximation algorithms. Optional enrichment problems are listed on the course page; these are not collected or graded but strongly recommended as exam preparation.
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## APPENDIX B: RESOURCES

Textbook: Introduction to Algorithms, 4th edition. Supplementary lecture videos are linked from the course page. The course staff maintain a curated list of practice problems sorted by topic and difficulty.

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